Metabolic Cholestasis, Part One: Galactosemia, Tyrosinemia Type 1 and Hereditary Fructose Intolerance in the Jaundiced Infant.

Picture three infants who arrive on the same ward in the same month. The first is a six-day-old girl, exclusively breastfed, who has become jaundiced, sleepy and feeds poorly. Her blood culture grows Escherichia coli, the common gut bacterium usually shortened to E. coli, and everyone is satisfied that sepsis explains the whole picture. The second is a seven-week-old boy whose international normalized ratio, the clotting test usually shortened to INR, is four, even though his bilirubin is only modestly raised and his liver enzymes are not dramatic. His newborn screen was reported as normal. The third is a five-month-old boy who was well on breast milk, began vomiting and sweating a few weeks after his parents started fruit purees, and whose blood sugar on admission is low. All three have an inborn error of metabolism. All three can die of it. And in all three the treatment is decided in large part by what the child is fed.

The clinical question of this chapter is how to recognize the three classic metabolic liver diseases of early infancy, galactosemia, tyrosinemia type 1 and hereditary fructose intolerance, before they declare themselves through sepsis, liver failure or hypoglycemia. The main idea is this. These diseases rarely announce themselves through bilirubin alone. They announce themselves through synthetic failure of the liver, through infection, or through low blood sugar. Two of them, galactosemia and hereditary fructose intolerance, need a particular sugar that the infant has to be fed. The third, tyrosinemia, needs only ordinary protein and so does not wait for any particular food. That single fact, the feeding history, organizes almost everything that follows, including the answer to a question students often ask: why hereditary fructose intolerance never appears in an infant who is fed only breast milk.

We will move through the topic in order. First, the mechanism: three blocked enzymes and three different poisons. Second, how often these diseases appear among infants with cholestasis and liver failure. Third, galactosemia, including its striking link with E. coli sepsis, the right way to test for it, and what happens in the long term. Fourth, tyrosinemia type 1, its newborn screening blind spot, and the remarkable effect of the drug nitisinone. Fifth, hereditary fructose intolerance and why breast milk protects. We then put the three side by side, describe a practical sequence for the first twenty-four hours, compare the guidelines, list the traps, and finish with what to do differently in clinic.

A note on sources. For the cholestatic infant in general we use the 2017 joint guideline on the evaluation of cholestatic jaundice in infants, written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as

NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. For galactosemia we use the 2017 international clinical guideline of the Galactosemia Network, known as GalNet, published by Welling and colleagues, together with the GeneReviews chapter by Berry, last updated in 2021. For tyrosinemia we use the 2017 United States and Canada consensus recommendations by Chinsky and colleagues, and the Quebec cohort reported by Larochelle and colleagues in 2012. For hereditary fructose intolerance we use the GeneReviews chapter by Gaughan, Ayres and Baker, updated in 2021. For liver failure in young infants we use the 2011 report of the Pediatric Acute Liver Failure Study Group by Sundaram and colleagues.

Part one. Three blocked enzymes, three different poisons.

A useful rule for all three diseases is that the harm comes from what accumulates above the blocked step, not from the lack of what lies below it. Each disease is a recessive deficiency of one enzyme, and in each the substance stuck in front of the block is toxic.

Start with galactosemia. Lactose, the sugar of all mammalian milk, is split in the gut into glucose and galactose. Galactose is handled in the liver by a short pathway, and the critical enzyme is galactose-1-phosphate uridylyltransferase, mercifully shortened to GALT. When GALT is absent, galactose-1-phosphate builds up inside cells of the liver, the kidney tubules and the red blood cells, and it injures them. At the same time, some galactose is diverted to a sugar alcohol called galactitol. Galactitol cannot leave the cell. In the lens of the eye, galactitol pulls water in by osmosis, the lens fibers swell, and a cataract forms. So the galactosemic infant can have a sick liver, leaky kidney tubules and cloudy lenses, all within the first weeks of feeding.

Tyrosinemia type 1 is a disease of amino acid breakdown. The enzyme missing is fumarylacetoacetate hydrolase, shortened to FAH, the last step in the breakdown of the amino acid tyrosine. The compounds that build up in front of the block, fumarylacetoacetate and its relatives, are reactive and damage the hepatocyte and the kidney tubule cell directly. One of their products, succinylacetone, is the diagnostic marker of the disease. Succinylacetone also blocks an early step in the making of heme, which explains the episodes of pain, high blood pressure and nerve dysfunction that resemble acute porphyria. An important clinical consequence is that in tyrosinemia the liver tends to lose its ability to make clotting proteins before it loses its ability to excrete bilirubin. The INR can therefore be grossly prolonged while the bilirubin and the transaminases are only modestly raised.

Hereditary fructose intolerance is caused by deficiency of aldolase B, the liver enzyme that splits fructose-1-phosphate. When an affected child eats fructose, or sucrose, which is half fructose, or sorbitol, which the body turns into fructose, the liver quickly phosphorylates the

fructose to fructose-1-phosphate and then cannot split it. Fructose-1-phosphate accumulates and traps the cell's inorganic phosphate. Without free phosphate, the cell cannot regenerate ATP, the energy currency of the cell. As ATP is broken down, uric acid rises. Glycogen breakdown and the production of new glucose both stall, so the blood sugar falls, and glucagon may fail to raise it. Repeated exposure leads to an enlarged, fatty and eventually scarred liver and to damage of the proximal kidney tubule.

Now we can answer the question of the breastfed infant. The only sugar in human milk is lactose, which is glucose joined to galactose. There is no fructose, no sucrose and no sorbitol in breast milk. An infant with hereditary fructose intolerance who is fed only breast milk never needs aldolase B and so stays entirely well. The disease appears when fructose arrives: at weaning, with fruit and juice, with a formula sweetened with sucrose, with sweetened medicines, or, in the newborn unit, with concentrated sucrose given for procedural pain. By contrast, the galactosemic infant is exposed from the very first feed, whether that feed is breast milk or a standard formula, because both contain lactose.

Part two. How often these diseases appear.

These are rare disorders, but they matter out of proportion to their frequency because they are treatable and because they cause liver failure rather than mere jaundice. The 2017 joint guideline estimates that cholestatic jaundice affects about one in two thousand five hundred term infants. A 2015 systematic review by Gottesman and colleagues, which pooled one thousand six hundred and ninety-two cholestatic infants, found metabolic disease in about four point four percent, and galactosemia was the single most common metabolic cause, accounting for about thirty-seven percent of the metabolic cases.

The picture becomes sharper when we look at liver failure rather than cholestasis. In the Pediatric Acute Liver Failure Study Group report by Sundaram and colleagues, one hundred and forty-eight infants aged ninety days or younger had acute liver failure. Twenty-eight of them, almost nineteen percent, had a metabolic cause. When the slide drew these causes as bars, one bar dominated: galactosemia, with twelve infants, about eight percent of the whole cohort. Respiratory chain defects followed with five, and then tyrosinemia, Niemann-Pick disease type C and other mitochondrial disorders with three each, with single cases of urea cycle defects. Overall, sixty percent of these young infants survived without transplant. The message is simple: in a very young infant whose liver is failing, galactosemia is the metabolic diagnosis to exclude first, and tyrosinemia is next.

Part three. Galactosemia.

The natural history of untreated classic galactosemia unfolds over about a week. An infant starts breast milk or a lactose-containing formula from the first day. Within days come poor

feeding and vomiting, then jaundice, an enlarged liver and bleeding from coagulopathy. The kidney tubules leak, and cataracts may form. In the GeneReviews summary of three hundred and fourteen individuals, cataract was found in thirty percent, but about half of these were mild or transient and resolved on diet, and only eight needed surgery. Then comes the event that kills: sepsis, typically with E. coli. And yet, when lactose is removed from the diet within the first three to ten days, the neonatal signs settle quickly.

The link with E. coli deserves its own attention, because it is one of the most useful pattern-recognition clues in neonatal hepatology. In 1977, Harvey Levy and colleagues reported the experience of the Massachusetts newborn screening program, in a short abstract and a companion paper in the New England Journal of Medicine. Of ten galactosemic infants identified, four developed severe E. coli infection, and three of the four died of bacteremia or meningitis. All three deaths occurred in infants whose diagnosis came after the first week of life. The seven infants treated before the sixth day did well. The authors found normal immunoglobulin levels, and in their laboratory, galactose at clinically relevant concentrations did not clearly impair the killing of E. coli by white cells, so the mechanism of this susceptibility remains unsettled. A larger international survey published by Waggoner and colleagues in 1990 found sepsis in about ten percent of two hundred and seventy symptomatic newborns, with E. coli in twenty-six. GeneReviews makes the essential practical point: sepsis does not exclude galactosemia. The bedside rule follows directly. A jaundiced newborn with E. coli sepsis should have lactose stopped and a red cell GALT sample sent before any blood transfusion.

That leads to the first and strongest recommendation of the 2017 GalNet guideline. When classic galactosemia is suspected, a galactose-free formula should be started immediately, without waiting for confirmation. Suitable choices are a soy formula, a casein hydrolysate or an elemental formula. Breastfeeding and expressed breast milk are contraindicated in classic galactosemia, as the United States Centers for Disease Control and Prevention also states. The cost-benefit is lopsided. A few days of hydrolysate in an infant who turns out not to have galactosemia costs very little. One more day of lactose in an infant who does have it can mean sepsis or liver failure.

Testing has a few traps that every hepatologist should know. The definitive test is the activity of GALT in red blood cells. Absent activity, or activity below about ten percent, defines classic galactosemia, and GalNet recommends treatment at that level or when two disease-causing variants of the GALT gene are found, including a variant called S135L. The trap is transfusion. Donor red cells contain normal GALT, so a sample drawn after a transfusion, which is common in a bleeding, septic newborn, can give a falsely normal result for weeks. This is why the sample must be sent first. If a transfusion has already been given,

the fallbacks are urinary galactitol, which GeneReviews describes as especially useful in this situation, and genetic testing of GALT, which takes days but does not depend on the red cells. Red cell galactose-1-phosphate is high at diagnosis and is used for monitoring, at three and nine months according to GalNet, but it falls once lactose is stopped. Urine reducing substances remain a tier one test in the 2017 joint guideline, but in expert practice a negative result means little in an infant who is not actually taking lactose, for example one who has been on intravenous fluids. Newborn screening for galactosemia has been universal in the United States since 2004, when all fifty-one programs screened, and classic galactosemia is found in roughly one in forty-eight thousand births. The difficulty is timing: the screening result may arrive after the window in which sepsis develops.

One more category deserves mention because it causes needless anxiety. The Duarte variant, which combines one classic allele with one Duarte allele, is found in about one in four thousand white infants. These infants have partial GALT activity, commonly around a quarter of normal. GalNet is clear: do not treat Duarte variant galactosemia, and no endocrine follow-up is needed because the ovaries are not affected.

The long-term story is sobering. The diet saves the liver, but it does not reliably protect the brain or the ovary. The slide showed this as a set of bars. Primary ovarian insufficiency affected eighty-one percent of forty-seven girls and women. Speech problems affected fifty-six percent of two hundred and forty-three individuals, and only about a quarter of these resolved; some had childhood apraxia of speech. Developmental delay was found in forty-five percent of one hundred and seventy-seven individuals aged six or older, and tremor or coordination problems in eighteen percent of those older than five. A 2011 review by Fridovich-Keil and colleagues put ovarian insufficiency at more than eighty and perhaps more than ninety percent of females with classic galactosemia. Perhaps most surprising, the 1990 international survey found that poor long-term outcome was not explained by late diagnosis or by poor diet control. Something about the disease continues despite the diet; one leading explanation is that the body makes a small amount of galactose itself. GalNet therefore recommends structured follow-up: intelligence testing from age two to three onward, speech screening from seven to twelve months, review of puberty by age twelve, and bone density assessment from about eight to ten years.

Part four. Tyrosinemia type 1.

Tyrosinemia type 1 produces a liver that fails before it turns yellow. The 2017 consensus emphasizes that coagulopathy is an early sign and can appear without other striking liver abnormalities, and the 2017 joint cholestasis guideline notes that these infants may present with coagulopathy out of proportion to their other abnormalities. Alpha-fetoprotein, the fetal protein usually shortened to AFP, is usually very high, but normal newborns also have high

AFP, so a single high value is not diagnostic on its own; the trend matters more. The kidney is the second target, with Fanconi syndrome, which is a generalized leak of the proximal tubule, causing hypophosphatemic rickets, acidosis and loss of amino acids in the urine. The nerves are the third target, with porphyria-like crises of pain, high blood pressure and low sodium that can progress to respiratory failure. The fourth target is cancer. Before nitisinone, hepatocellular carcinoma developed in between seventeen and thirty-seven percent of affected children. And infants who presented with liver failure before two months of age had a two-year survival of only twenty-nine percent in the pre-nitisinone era, as reported by van Spronsen and colleagues in 1994.

The newborn screening blind spot is the most important practical point in this section. Tyrosine itself is a poor screening marker. In a 2014 analysis by De Jesús and colleagues for the United States Centers for Disease Control, at least half of affected infants had screening tyrosine levels that overlapped the normal range, and the 2017 consensus estimated that a tyrosine-first approach may miss about twenty-eight percent of patients. Many affected infants simply have normal tyrosine when sampled before forty-eight hours of age. In 2014, fifty of fifty-one United States programs screened for tyrosinemia type 1, but only thirty-eight measured succinylacetone. The consensus therefore recommends blood succinylacetone as the primary screening marker, and advises against tyrosine as the primary marker. The practical lesson is to ask which marker your state or country uses. A normal screen from a tyrosine-based program does not exclude tyrosinemia in a cholestatic infant with coagulopathy. Send urine or blood succinylacetone.

The disease is unusually common in one region of Quebec, the Saguenay to Lac-Saint-Jean area, where it affects about one in one thousand eight hundred and forty-six births, compared with about one in one hundred thousand worldwide. That founder population allowed one of the clearest natural experiments in pediatric hepatology. Larochelle and colleagues followed seventy-eight patients born between 1984 and 2004 in three groups: never treated with nitisinone, treated after one month of age, and treated before one month. The slide showed the proportion needing liver transplantation as three falling bars: seventy-one percent of the untreated, twenty-six percent of the late-treated, and none of the early-treated. There were one hundred and eighty-four hospitalizations for acute complications during untreated periods and none on nitisinone. Ten untreated patients died, compared with two treated patients, and both treated deaths were complications of transplantation. No early-treated patient had detectable liver disease after more than five years. Nitisinone itself was first used by Lindstedt and colleagues in 1992; in their first five patients, succinylacetone excretion fell within hours. The drug works by blocking an earlier enzyme in tyrosine breakdown, so that the toxic compounds are never made, at the price of a higher tyrosine level that must be controlled with a low-protein diet.

The timing of treatment also governs cancer risk. Data cited in the 2017 consensus showed a gradient: when nitisinone was started before one year of age the risk of hepatocellular carcinoma was under one percent, rising to seven percent when started at one to two years, twenty-one percent at two to seven years, and thirty-five percent when started after seven years. The consensus treatment targets are worth memorizing. Start at one milligram per kilogram per day in two doses, increasing to two milligrams per kilogram in acute liver failure or poor response. Aim for a blood nitisinone level of forty to sixty micromoles per liter with undetectable succinylacetone, and keep plasma tyrosine between two hundred and six hundred and phenylalanine between twenty and eighty micromoles per liter on a low-protein diet. AFP should fall steadily through the first year; a plateau or a rise is a reason to image the liver at once. The liver should be imaged every three to six months. Transplantation should be considered if there is no response after about a week, or after two to three days in an infant with severe coagulopathy or encephalopathy.

Part five. Hereditary fructose intolerance.

We have already seen why the breastfed infant is safe. The sequence when fructose arrives is predictable. Weaning foods, juice, a sucrose-sweetened formula, sorbitol in a syrup or sucrose for analgesia provide the exposure. Fructose-1-phosphate accumulates and traps phosphate, ATP falls and uric acid rises. Glycogen breakdown and new glucose production stall, producing hypoglycemia that may not respond to glucagon. Repeated exposure produces hepatomegaly, steatosis, fibrosis and sometimes liver failure, together with proximal tubulopathy. Two neonatal exceptions are worth remembering. GeneReviews records deaths from hypoglycemia after twenty-four percent sucrose was given for procedural pain. And the oral rotavirus vaccines, Rotarix and RotaTeq, each contain more than one thousand milligrams of sucrose per dose and are avoided in a child known to have the disease. In the older child, the history is often the test: a natural aversion to sweets and fruit, notably healthy teeth, and unexplained steatosis or tubular dysfunction.

The disease is not as rare as its reputation. Estimates based on carrier frequencies are about one in eighteen thousand in the United Kingdom, one in twenty-six thousand in Germany, and one in thirty-one thousand in Poland. In a British newborn study by James and colleagues, one point three two percent of newborns, twenty-seven of two thousand and fifty, carried the A149P variant, the commonest European variant. Diagnosis today rests on sequencing of the aldolase B gene, which detects two disease-causing variants in seventy-five to one hundred percent of affected people. Liver biopsy enzyme assay is now rarely needed. One test must never be used: an intravenous or oral fructose tolerance test, which has caused deaths. Treatment is lifelong exclusion of fructose, sucrose, sorbitol and the sweetener sucralose, with a sugar-free multivitamin. With adherence, the prognosis is excellent and life expectancy normal. It is worth noting where hepatology meets this disease: the 2017 cholestasis guideline

mentions hereditary fructose intolerance only as an association with the feeding source and names no specific test. The clinician has to ask the feeding question.

Part six. Putting the three together.

At the bedside, the three diseases separate cleanly along a few lines. The trigger for galactosemia is lactose from the first feed; tyrosinemia needs no particular trigger because protein is always present; hereditary fructose intolerance needs fructose, sucrose or sorbitol. Galactosemia presents in days three to ten, tyrosinemia over weeks to months, and fructose intolerance at weaning or with neonatal sucrose. The signature of galactosemia is E. coli sepsis and cataract, of tyrosinemia an INR out of proportion to jaundice together with rickets, and of fructose intolerance hypoglycemia after sweet feeds. The key test is red cell GALT, sent before transfusion, for galactosemia; succinylacetone in urine or blood for tyrosinemia; and aldolase B sequencing for fructose intolerance. Newborn screening covers galactosemia in all United States programs, covers tyrosinemia reliably only where succinylacetone is measured, and does not cover fructose intolerance at all. Treatment is lifelong lactose restriction, nitisinone with a low-protein diet, and lifelong fructose restriction, respectively.

In practice, the first twenty-four hours with a sick cholestatic infant can follow a simple sequence. First, recognize the warning signs: an INR that stays high despite vitamin K, Gram-negative sepsis, low glucose or cataract. Second, sample before you treat: red cell GALT before any transfusion, urine and blood succinylacetone, AFP, lactate and ammonia. Third, change the feed to a formula that is both lactose-free and sucrose-free, such as an extensively hydrolyzed or elemental formula, and give intravenous glucose. Fourth, treat: antibiotics, vitamin K, and nitisinone as soon as succinylacetone is found to be high. Fifth, refer early to a liver transplant center if liver failure progresses. The reason to insist on a sucrose-free formula is that a switch to a lactose-free formula sweetened with sucrose protects against galactosemia while exposing an infant with fructose intolerance. Read the label.

Part seven. What the guidelines say.

Comparing North American and European documents is simpler here than for most topics, because the main pediatric hepatology guideline is shared and neither the American Association for the Study of Liver Diseases nor the European Association for the Study of the Liver has specific guidance on these disorders in infants. For testing for galactosemia in the cholestatic infant, the 2017 joint guideline places urine reducing substances in tier one and red cell GALT in tier two, and GalNet adds GALT genotyping; the verdict is aligned. For diet on suspicion, the Centers for Disease Control state that breast milk is contraindicated in classic galactosemia, and GalNet gives a strong recommendation for an immediate galactose-free formula; again aligned. For the Duarte variant, GalNet says do not treat, while there is no comparable North American society statement; the verdict is that the two differ. For the

tyrosinemia screening marker, the United States and Canada consensus insists on succinylacetone, yet in 2014 only thirty-eight of fifty-one programs measured it, and there is no ESPGHAN or EASL statement; this is a material difference, because a cholestatic infant can carry a normal tyrosinemia screen if the program measures only tyrosine. For hereditary fructose intolerance, the joint guideline offers only the feeding history, and GeneReviews points to sequencing; this is a gap.

Part eight. Traps, and what to do differently in clinic.

Four traps account for most late diagnoses. The first is the phrase "just sepsis": E. coli seems to explain the illness, lactose is continued, and the diagnosis comes after the first week, exactly the pattern of the deaths in the Massachusetts cohort. The second is "the screen was normal": the program may have measured tyrosine rather than succinylacetone, the sample may have been taken after a transfusion, or the result may simply not be back yet. The third is "AFP is high anyway": newborn AFP is high, but the trend matters, and the INR is the better clue. The fourth is "a new formula": a lactose-free but sucrose-sweetened formula unmasks fructose intolerance. The common thread is that a septic, jaundiced or bleeding newborn deserves a metabolic question on the first day, not after the cultures come back.

To bring these lessons to the ward, consider five short scenarios. A six-day-old breastfed girl who is jaundiced with E. coli in her blood should be thought of as having galactosemia: stop breast milk, start a hydrolysate, send red cell GALT and ask for an eye examination. A seven-week-old with an INR of four, a bilirubin of four milligrams per deciliter and a very high AFP should be thought of as having tyrosinemia: send urine and blood succinylacetone, give vitamin K, and start nitisinone if the result is positive. A five-month-old with vomiting and low glucose after starting fruit has fructose intolerance until proven otherwise: remove fructose, sucrose and sorbitol and send aldolase B sequencing. A cholestatic infant transfused before GALT was sent still has unresolved galactosemia: send urinary galactitol and GALT genotyping, and keep the feeds lactose-free. And an infant with a positive galactosemia screen, GALT activity around twenty-five percent and a Duarte genotype needs no diet and no endocrine follow-up.

The whole chapter can be held in two sentences. In a sick cholestatic infant, take the feeding history first, remove lactose and sucrose while the tests run, and send red cell GALT before any transfusion and succinylacetone rather than tyrosine. Treat coagulopathy out of proportion to jaundice as a diagnosis, start nitisinone early if succinylacetone is high, and remember that fructose intolerance cannot appear while the infant takes only breast milk.

Questions to test yourself.

Question one. Why does a galactosemic infant develop cataracts? Answer: galactose is reduced to galactitol, which cannot leave the lens cell, draws in water by osmosis and clouds the lens.

Question two. What happened to the galactosemic infants in the 1977 Massachusetts cohort? Answer: four of ten developed severe E. coli infection and three died, all of them diagnosed after the first week of life, while the seven treated before day six did well.

Question three. Why must red cell GALT be sent before transfusion? Answer: donor red cells contain normal GALT and can make a deficient infant look normal for weeks; afterwards, urinary galactitol and genotyping are the fallbacks.

Question four. Does early diet prevent the late complications of galactosemia? Answer: not reliably; ovarian insufficiency affects more than eighty percent of females and speech problems more than half of patients, and outcome was not explained by late diagnosis in the 1990 survey.

Question five. Why can a newborn screen miss tyrosinemia type 1? Answer: at least half of affected infants have screening tyrosine in the normal range, so programs must measure succinylacetone; in 2014 only thirty-eight of fifty-one United States programs did.

Question six. What did the Quebec cohort show about the timing of nitisinone? Answer: liver transplantation was needed in seventy-one percent of untreated patients, twenty-six percent of those treated after one month, and none of those treated before one month.

Question seven. Why does hereditary fructose intolerance not occur in an exclusively breastfed infant? Answer: the only sugar in breast milk is lactose, made of glucose and galactose, so aldolase B is never needed until fructose, sucrose or sorbitol is introduced.

End of chapter.